to test Rails functionality. We'll also talk a bit about what makes testing and tests most valuable.

After that we'll spend a few chapters going through basic unit tests, talking about models, multiple ways of generating data for tests, and using test doubles to simulate objects and specify hard-to-reach states.

Then we'll zoom out a little bit and talk about end-to-end tests. We'll also start talking about JavaScript, first as an extension to end-to-end tests for code that has JavaScript, and then about unit-testing JavaScript by itself. Then we'll tour the remaining parts of Rails that we haven't covered and show the tool support for those systems.

Once we've gone though all that, we'll walk through most of it again using Minitest instead of RSpec.

Once we're done talking about tools, we'll cover specific scenarios for testing, including testing for security and testing third-party services. We'll talk about troubleshooting failing tests and how to improve your test environment and run your tests quickly. And we'll end with the very common case where you need to add tests to untested legacy code.

What You'll Need

The sample code and mini application you build in this book use Ruby 2.5.0 and Rails 5.2.0,[1] [2] which are the current versions of these tools as I write. (Older versions will run most of the book's examples.) Installation of these tools is beyond the scope of the book, but you can find instructions at their footnoted websites.

The application also uses the SQLite database. I chose SQLite for ease of installation. You may already have it installed on your system; if not, you can find installation instructions on the SQLite website.[3]

A Word About Tools, Best Practices, and Teaching TDD

Two test libraries are in general use in the Rails community—Minitest and RSpec—which means I had a choice of which tool to use as the primary library in this book. This book is about how to test Rails in general; therefore, the details of the testing library in use are secondary to most of it. Still, the examples have to be presented using one tool or the other.

1. https://www.ruby-lang.org
2. http://www.rubyonrails.com
3. https://sqlite.org